

GAURAV AHIR

Embedded Systems Engineer

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SUMMARY

Embedded Systems Engineer with hands-on experience in STM32, ESP32, PIC18, SIM800L, and QUECTEL-EC200U based firmware development. Proficient in real-time C firmware, UART/SPI/I2C communication, AT command stacks, GSM integration, and low-power IoT design. Experienced in delivering production-ready firmware across the full product lifecycle, from R&D and component selection through PCB design, QC, and field deployment.

SKILLS

Languages: Embedded C, C++, Python, Assembly

Microcontrollers & Platforms: STM32, ESP32, PIC18, Arduino

Communication Protocols: UART, I2C, SPI, RS485, Modbus, AT Commands, MQTT, HTTP

Tools & IDEs: STM32CubeIDE, ESP-IDF, MPLAB IDE, VS Code, Arduino IDE, Git

Firmware Development: HAL/LL Drivers, Interrupt Handling, ISR, ADC, PWM, Timers, Watchdog Timer, GPIO, Sleep Modes, Low-Power Design, Bootloader, OTA Updates

RTOS & Scheduling: FreeRTOS, Multithreading, Task Scheduling, Buffer Handling, State Machine Design

Debugging & Testing: Serial Monitor (USART), Logic Analyzer, AT Command Testing

Hardware Design: Schematic Design, PCB Layout (up to 4-Layer), RF Design, High-Speed Signal Routing, Thermal & Ground Plane Design

PCB Design Software: Altium Designer, KiCad

EXPERIENCE

Jr. Embedded Engineer

1 Yr 1 Mo (2025 – Present)

Independent Project Development

R&D, Firmware and Hardware, Full Product Lifecycle

- SNI Automation - Deployed 3 embedded devices from scratch; reduced power consumption, developed fast-responsive firmware with error handling.

Freelance Project Work

6 Mo (2026 – Present)

Independent Project Development

Altium Designer, Datasheet Reading

- Successfully deployed 20+ embedded systems projects, including RF, high-speed, and 4-layer PCB designs.

PROJECTS

Perimeter Intrusion Detection System

- LoRa-based three-layer centralized advanced perimeter intrusion detection system.
- UART, I2C, and SPI protocol-based sensors, AI/ML-based Camera modules, and wireless communication using LoRa Protocol, 9 Axis Gyro Sensor, mm-Wave Motion detection Sensor.
- Developed firmware for sensor data processing, event flag generation, and FIFO-based data transmission with ISR-controlled reception.
- **Demonstrated project at DRDO Bhavan Delhi, Jaisalmer Army Camp, and Gandhinagar Army Camp.**

Railway Water Tank Level Measurement IoT Device

- 4G LTE IoT device using QUECTEL-EC200U modem with AT command stack; transmitted tank level, GPS coordinates, and battery telemetry to CRIS server via MQTT over TCP/IP.
- Interfaced 4-20mA current loop sensor via ADC with calibrated scaling for accurate water level measurement; additional I2C and UART peripheral integration.
- Implemented low-power sleep-wake firmware cycle, reducing average current consumption from 200mA to 40mA — an **80% reduction** in idle-state power draw.
- Managed full project lifecycle: tender document review, component selection, schematic design, 4-layer PCB layout, QC, and production deployment.

Payment Announcer Device

- STM32-based device using SIM800L GSM module; firmware written in C using STM32CubeIDE.
- Implemented interrupt-driven UART reception with circular FIFO buffer for SMS parsing and audio amplification.
- Designed non-blocking state machine for GSM health monitoring and network recovery.

EDUCATION

B.E. Instrumentation & Control

GEC Gandhinagar, Gujarat

Graduated: 2025

CGPA: 6.8